Link to the Colab Notebook: https://colab.research.google.com/drive/1JBiCBgDs9P8VYQcgKk8xvjVSg1fxArGS?usp=sharing

Contributions

Yukun Wang: Prepared presentation slides and model training in Part 2.
Patuan Purba: Model training in Part 2 and work in Part 1.
Jack Li: Authored this report and one model training in Part 2.

Part 1

A.

By counting the number of occurrences of each label in the column *Sentiment* and dividing them by the total number of observations, we obtain

Sentiment	2	0	1
balance	0.438434	0.374159	0.187407

Figure 1: Balance of Each Class In The Training Set

B.

By using $nltk.word_tokenize$, we obtain

	OriginalTweet	Tokenized_tweets	Sentiment
0	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	[@, MeNyrbie, @, Phil_Gahan, @, Chrisitv, http...	1
1	advice Talk to your neighbours family to excha...	[advice, Talk, to, your, neighbours, family, t...	2
2	Coronavirus Australia: Woolworths to give elde...	[Coronavirus, Australia, :, Woolworths, to, gi...	2
3	My food stock is not the only one which is emp...	[My, food, stock, is, not, the, only, one, whi...	2
4	Me, ready to go at supermarket during the #COV...	[Me, ,, ready, to, go, at, supermarket, during...	0
...	...	...	...
41150	Airline pilots offering to stock supermarket s...	[Airline, pilots, offering, to, stock, superma...	1
41151	Response to complaint not provided citing COVI...	[Response, to, complaint, not, provided, citin...	0
41152	You know itÃ s getting tough when @KameronWild...	[You, know, itÃ s, getting, tough, when, @, Ka...	2
41153	Is it wrong that the smell of hand sanitizer i...	[Is, it, wrong, that, the, smell, of, hand, sa...	1
41154	@TartiiCat Well new/used Rift S are going for ...	[@, TartiiCat, Well, new/used, Rift, S, are, g...	0

41151 rows × 3 columns

Figure 2: Training Set with Tokenized Tweets

C.

Using the regular expression `^http\S*` and removing any token that matches this pattern, we obtain

	OriginalTweet	Tokenized_tweets	Sentiment
0	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	[@, MeNyrbie, @, Phil_Gahan, @, Chrisitv, :, /...	1
1	advice Talk to your neighbours family to excha...	[advice, Talk, to, your, neighbours, family, t...	2
2	Coronavirus Australia: Woolworths to give elde...	[Coronavirus, Australia, :, Woolworths, to, gi...	2
3	My food stock is not the only one which is emp...	[My, food, stock, is, not, the, only, one, whi...	2
4	Me, ready to go at supermarket during the #COV...	[Me, ,, ready, to, go, at, supermarket, during...	0
...	...	...	...
41150	Airline pilots offering to stock supermarket s...	[Airline, pilots, offering, to, stock, superma...	1
41151	Response to complaint not provided citing COVI...	[Response, to, complaint, not, provided, citin...	0
41152	You know itÂ s getting tough when @KameronWild...	[You, know, itÂ s, getting, tough, when, @, Ka...	2
41153	Is it wrong that the smell of hand sanitizer i...	[Is, it, wrong, that, the, smell, of, hand, sa...	1
41154	@TartiiCat Well new/used Rift S are going for ...	[@, TartiiCat, Well, new/used, Rift, S, are, g...	0

41151 rows × 3 columns

Figure 3: Training Set with Tokenized URL Removed Tweets

D.

By defining the regular expression [`^\w\s`] to remove every punctuation or special character in the tokens and extracting the lowercase of the resulting tokens, we obtain

	OriginalTweet	Tokenized_tweets	Sentiment
0	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	[menyrbie, phil_gahan, chrisitv, tcoifz9fan2pa...	1
1	advice Talk to your neighbours family to excha...	[advice, talk, to, your, neighbours, family, t...	2
2	Coronavirus Australia: Woolworths to give elde...	[coronavirus, australia, woolworths, to, give,...	2
3	My food stock is not the only one which is emp...	[my, food, stock, is, not, the, only, one, whi...	2
4	Me, ready to go at supermarket during the #COV...	[me, ready, to, go, at, supermarket, during, t...	0
...	...	...	...
41150	Airline pilots offering to stock supermarket s...	[airline, pilots, offering, to, stock, superma...	1
41151	Response to complaint not provided citing COVI...	[response, to, complaint, not, provided, citin...	0
41152	You know itÂ s getting tough when @KameronWild...	[you, know, itâs, getting, tough, when, kamero...	2
41153	Is it wrong that the smell of hand sanitizer i...	[is, it, wrong, that, the, smell, of, hand, sa...	1
41154	@TartiiCat Well new/used Rift S are going for ...	[tartiicat, well, newused, rift, s, are, going...	0

41151 rows × 3 columns

Figure 4: Training Set with Tokenized Lowercase Tweets With URL, Special Characters, Punctuations Removed

E.

By using a Porter Stemmer to stem each tokenized tweet, we obtain

	OriginalTweet	Tokenized_tweets	Sentiment
0	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	[menyrbi, phil_gahan, chrisitv, tcoifz9fan2pa,...	1
1	advice Talk to your neighbours family to excha...	[advic, talk, to, your, neighbour, famili, to,...	2
2	Coronavirus Australia: Woolworths to give elde...	[coronaviru, australia, woolworth, to, give, e...	2
3	My food stock is not the only one which is emp...	[my, food, stock, is, not, the, onli, one, whi...	2
4	Me, ready to go at supermarket during the #COV...	[me, readi, to, go, at, supermarket, dure, the...	0
...	...	...	...
41150	Airline pilots offering to stock supermarket s...	[airlin, pilot, offer, to, stock, supermarket,...	1
41151	Response to complaint not provided citing COVI...	[respons, to, complaint, not, provid, cite, co...	0
41152	You know itÂ s getting tough when @KameronWild...	[you, know, itâ, get, tough, when, kameronwild...	2
41153	Is it wrong that the smell of hand sanitizer i...	[is, it, wrong, that, the, smell, of, hand, sa...	1
41154	@TartiiCat Well new/used Rift S are going for ...	[tartiicat, well, newus, rift, s, are, go, for...	0

41151 rows × 3 columns

Figure 5: Training Set with Tokenized, Stemmed Lowercase Tweets With URL, Special Characters, Punctuations Removed

F.

After removing the first 100 stopwords from the list of *nltk*, we obtain

	OriginalTweet	Tokenized_tweets	Sentiment
0	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/l...	[menyrbi, phil_gahan, chrisitv, tcoifz9fan2pa,...	1
1	advice Talk to your neighbours family to excha...	[advic, talk, to, your, neighbour, famili, to,...	2
2	Coronavirus Australia: Woolworths to give elde...	[coronaviru, australia, woolworth, to, give, e...	2
3	My food stock is not the only one which is emp...	[food, stock, not, the, onli, one, which, empt...	2
4	Me, ready to go at supermarket during the #COV...	[readi, to, go, supermarket, dure, the, covid1...	0
...	...	...	...
41150	Airline pilots offering to stock supermarket s...	[airlin, pilot, offer, to, stock, supermarket,...	1
41151	Response to complaint not provided citing COVI...	[respons, to, complaint, not, provid, cite, co...	0
41152	You know itÂ s getting tough when @KameronWild...	[you, know, itâ, get, tough, when, kameronwild...	2
41153	Is it wrong that the smell of hand sanitizer i...	[wrong, that, the, smell, of, hand, sanit, sta...	1
41154	@TartiiCat Well new/used Rift S are going for ...	[tartiicat, well, newus, rift, s, go, 70000, o...	0

41151 rows × 3 columns

Figure 6: Training Set with Tokenized, Stemmed Lowercase Tweets With URL, Special Characters, Punctuations, Stop Words Removed

G.

Converting the tokens into word count using a CountVectorizer we obtain

- Size of the matrix of count: 41151×74194
- Length of vocabulary: 74194

H.

In this first model, we use PorterStemmer on tokenized tweets and a CountVectorizer to vectorize the tweets. After that, we feed the vectorized input into the Multinomial Naive Bayes Model with the Laplace smoothing coefficient equal to 1. We obtained Training Accuracy $= 0.7879$, and Testing Accuracy $= 0.6743$. Furthermore, the 5 most probable words in each class with their counts are

Class	Word 1	Word 2	Word 3	Word 4	Word 5
0	the: 18301	to: 14920	of: 9078	coronavirus: 6737	covid19: 4610
1	the: 6320	to: 4762	of: 3812	coronavirus: 2889	covid19: 2566
2	the: 20120	to: 18651	of: 9539	coronavirus: 7511	covid19: 5681

Table 1: Top 5 Most Probable Words Per Class with Occurrence Counts

I.

The ROC curve is not suitable here because the usual ROC curve is mainly for binary classification; however, this is a multi-class classification problem. Nevertheless, if we change the class label from $\{0, 1, 2\}$ to $\{0,\ \text{not } 0\}, \{1,\ \text{not } 1\},\ \text{or } \{2,\ \text{not } 2\}$, then we will have a binary classification problem and hence ROC curve will be a suitable measure.

J.

In this model, we use PorterStemmer on tokenized tweets and the TfidfVectorizer to vectorize the tweets. After that, we feed the vectorized input into the Multinomial Naive Bayes Model with the Laplace smoothing coefficient equal to 1. Then

$$\text{Training Accuracy} = 0.7223, \text{ and Testing Accuracy} = 0.6222.$$

Furthermore, the 5 most probable words in each class with their counts are the same as the previous table; hence we are skipping the presentation of it.
Upon inspection, we see that the previous model, i.e. the model that used CountVectorizer to vectorize

the tweets has a higher accuracy on both training and the testing set.
In particular, for the second model,

- on the training set, its accuracy is 0.0656 lower than the first model, and
- on the testing set, its accuracy is 0.0521 lower than the first model.

K.

In this model, we use WordNet Lemmatization on tokenized tweets and the TfidfVectorizer to vectorize the tweets. After that, we feed the vectorized input into the Multinomial Naive Bayes Model with the Laplace smoothing coefficient equal to 1. Then

$$\text{Training Accuracy} = 0.7224, \text{ and Testing Accuracy} = 0.6214.$$

Upon inspection, we see that this model perform slightly better on the training set and slightly worse on the testing set than the previous model,(i.e. the model that used stemming and a TfidfVectorizer to vectorize the tweets). In particular, for the second model,

- on the training set, its accuracy is 0.0001 higher than the previous model(stemming model), and
- on the testing set, its accuracy is 0.0008 lower than the previous model(stemming model).

Bonus

It is a generative model because it explicitly models the joint probability $P(X \mid Y)$ and $P(Y)$. Then according to Bayes's Theorem

$$P(X, Y) = P(Y) \cdot P(X \mid Y)$$

it can approximates $P(X, Y)$ and subsequently $P(Y \mid X)$.
This is by definition a generative model.

Part 2

Problem description and motivation

E-commerce platforms such as Amazon and eBay have significantly grown in global influence. Typically, these platforms allow customers to select a specific category and then browse products classified accordingly. However, when sellers initially list products, they primarily provide titles and textual descriptions without explicitly assigning these categories. This scenario necessitates algorithms capable of automatically categorizing products based solely on textual information. Consequently, our research question arises:
Given a textual description of a product, can we accurately classify this product into its corresponding category?
Applying Natural Language Processing techniques combined with machine learning methods presents a viable solution for automated product classification. Nonetheless, accurately categorizing products into one class as a human would pose several challenges. Consider a sweater. This sweater can either be classified as a sweater, cloth, outerwear, etc, based on its description. At the same time, product descriptions often contain ambiguity which is another challenge to correctly making classifications.

Describe the Data

Since we have finalized our research topic, we selected the "Amazon Product Dataset 2020," sourced from Kaggle, which comprises Amazon sales data from the year 2020. The dataset includes 9,147 observations and 31 features. Although it has 31 features, many of them are missing labels for every observation, and hence, this severely limits the number of features that we can use.
To prepare the dataset for our NLP analysis, we aggregated three textual features—the product title, product specification, and "about product" description—into a single combined feature labeled "Textual Information." This combined textual column encapsulates all the necessary information required for our NLP processing and is the feature that we use to train models. However, the aggregated textual information inevitably includes irrelevant or redundant content. This is a limitation of our study.

Exploratory Data Analysis

To gain a deeper understanding of our dataset, we first examined the distribution of product categories. The categories given from the data set are majorly of the form "category 1 | category 2." Observing this, we initially aimed to predict only the first-level categories for each product, and as a result, we plotted the following graph, focusing on the balance of each category.

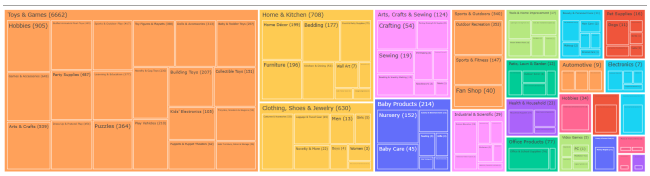

Figure 7: Tree Map of the Distribution of First Level Categories

However, as depicted in Figure 7, one category, "Toys & Games", dominated and accounts for approximately 66% of the dataset. Therefore, relying solely on these first-level categories for analysis could lead to highly biased predictions due to the overwhelming proportion of this single category.

To address this imbalance, for all category label whose first level is "Toys & Games", we decided to utilize their second-level categories. This approach significantly reduced category bias, resulting in a more balanced and representative distribution, as evident in our subsequent pie chart.

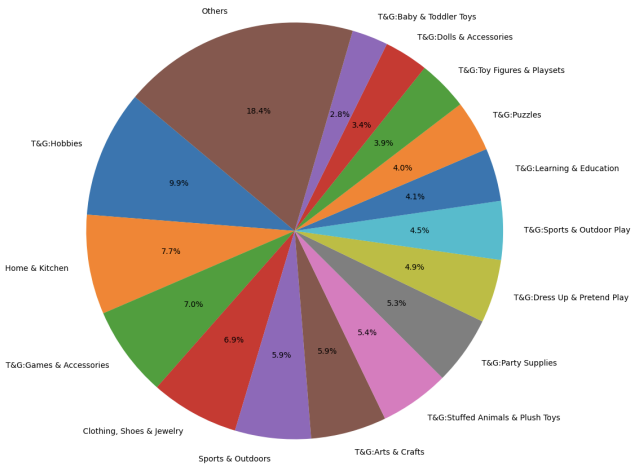

Figure 8: Proportion of Second-Level Categories (Top 15 with Others Grouped)

Before proceeding to NLP analysis, additional preprocessing steps were necessary to refine the textual data. We first tokenized the aggregated textual content, then applied regular expressions to clean and standardize the text. Subsequently, stemming techniques were used to normalize token variations, and records with missing values were dropped to ensure data consistency.
Furthermore, we conducted an analysis to identify the most frequently occurring words within the product descriptions. Interestingly, the most common terms identified were all related to children or kids. This finding aligns with our earlier observation regarding the prevalence of the "Games and Toys" category, reinforcing the notion that a significant portion of Amazon purchases are related to children, highlighting a potential driver behind consumer preference for online shopping.

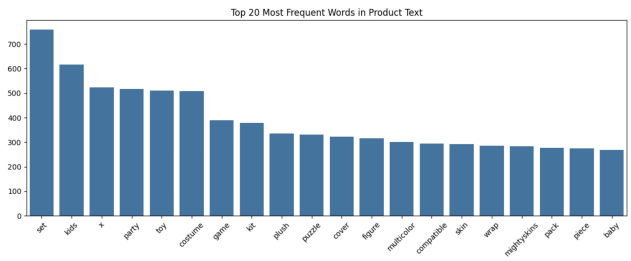

Figure 9: Top 20 Most Frequent Words in Product Text

For text vectorization, we selected different strategies depending on the machine learning model applied. In general, we employed both CountVectorizer and TF-IDF vectorizer, given their effectiveness in capturing textual features.

Describe Your Machine Learning Model

After completing the preprocessing and exploratory data analysis, we proceeded to the core phase of our project: training machine learning models to classify product categories based on textual data. We evaluated a total of five models — two introduced in class, one additional classic model, and two selected for extended exploration. These models include Naive Bayes, Logistic Regression, Linear

Support Vector Machine, Random Forest, and the more advanced Long Short-Term Memory network, which is a type of recurrent neural network. For model evaluation, we split the dataset into 60% training, 20% validation, and 20% testing subsets. All models employed in this study fall under the category of supervised learning, as they were trained using labeled product categories.

Model 1: Naive Bayes

This model calculates

$$P(y, \mid x_1, ..., x_n) = \frac{P(y) \cdot P(x_1, ..., x_n \mid y)}{P(x_1, ..., x_n)}$$

where $P(x_1, ..., x_n \mid y) = \prod_{i=1}^{n} P(x_i \mid y)$ which assumes all features are independent given the class. This is a relatively naive assumption, and hence the name "Naive" Bayes.

We consider this the simplest model in our study and treat it as our baseline model. The core idea behind this model is to calculate the likelihood of each class based on the frequency of words in the training data. After preprocessing, the cleaned and tokenized textual data is well-suited for this approach.

One of the primary advantages of this model is its speed and efficiency. However, it does not perform too well on the validation set, even when it was applied to clean, well-prepared text data, as seen below:

```
training_accuracy = accuracy_score(train_df['actual_category'], model_1.predict(counts_train))
validation_accuracy = accuracy_score(val_df['actual_category'], val_predictions)

print(f"Training Accuracy: {training_accuracy}")
print(f"Validation Accuracy: {validation_accuracy}")
print(classification_report(val_df['actual_category'], val_predictions, digits=3))

Training Accuracy: 0.7875704161366527
Validation Accuracy: 0.6864776444929117
```

Figure 10: Validation Accuracy for Model 1

We can see from the result that the validation accuracy for the data label is 0.686%. Though this is not a high score, it is a reasonable level of validation accuracy for a baseline model.

Model 2: Logistic Regression

In multi-class classification problems, for each class c, and input vector x, this model calculates a score

$$z_c = w_c^T \cdot x + b_c$$

where W_c is the weighted vector and b_c is the biased term for class C. The model then uses the softmax function, and transforms z_c into

$$\text{softmax}(z_c) = \frac{e^{z_c}}{\sum_{i=1}^{K} e^{z_i}}$$

where K is the number of total classes. This function effectively exponentializes each z_c so a higher score gets more weights, and normalizes the scores so all class probabilities sum to 1. The model classifies input x to class c if $\text{softmax}(z_c)$ is greatest among all other classes. This approach is well-suited to our dataset as we want to perform multi-class classification and have already vectorized each textual information, which this model requires, as discussed above. Below is the accuracy measure of this model.

The model performed well in our case. The validation accuracy reached 0.721, which marks a noticeable improvement over our baseline model, demonstrating Logistic Regression's effectiveness in handling textual classification tasks.

```python
# calculate accuracy for training and validating set
train_accuracy = accuracy_score(logisticmodel.predict(X_train), y_train)
val_accuracy = accuracy_score(logisticmodel.predict(X_val), y_val)

print('Accuracy on training set:', train_accuracy)
print('Accuracy on validating set:', val_accuracy)
```

```
Accuracy on training set: 0.831182991095766
Accuracy on validating set: 0.7153762268266085
```

Figure 11: Accuracy Measure of Model 2

Model 3: Linear SVM

This model aims to find the optimal hyperplane that best separates data points in a high-dimensional space. For binary classification, the model finds all possible (w, b) where w is the supporting vector and b is the biased term, such that $w^T x + b = 0$. In other words, (w, b) defines a the decision boundary(a hyperplane). Then for all such decision boundaries, the model maximizes the distance between the boundary and the closest data point from each class, while ensuring the boundary separates the classes. Based on this, we did a multinomial SVM. For each class, the model trained an SVM that classifies this class or not this class. Then the model classifies input x into the class that has the highest possibility of being classified into that class. Since our textual features are vectorized, SVM is well-suited for identifying distinct boundaries between product categories, making it an appropriate model choice for this problem.
However, while SVM performs effectively on smaller or moderately sized datasets, it can become computationally expensive and less interpretable as the dataset size grows.

```python
print("Training Accuracy:", train_accuracy)
print("Validation Accuracy:", val_accuracy)

print("\nTraining Classification Report:")
print(classification_report(y_train, y_train_pred, target_names=le.classes_))
```

```
Training Accuracy: 0.9847355987643104
Validation Accuracy: 0.7459105779716467
```

Figure 12: Accuracy Measure for Model 3

The model has an accuracy score of 0.98 on the training set yet an accuracy score of only 0.75 on the validation set. Given the model exhibits an unusually high training accuracy, we therefore concluded the model exhibits overfitting.

Model 4: Random Forest

Random Forest is an advanced ensemble algorithm composed of multiple decision trees. Each decision tree is trained on a random subset of the training set. Then given an input x, each decision tree makes a classification on x, and the class with the highest vote is the classification made by the model(or the forest). With a forest consisting of 100 decision trees, we have the following.

```python
# Report
print("Training Accuracy:", accuracy_score(y_train, y_train_pred))
print("Validation Accuracy:", accuracy_score(y_val, y_val_pred))
print(classification_report(y_val, y_val_pred, labels=valid_labels,
```

```
Training Accuracy: 0.9801008586615783
Validation Accuracy: 0.7579062159214831
```

Figure 13: Accuracy Measure for Model 4

Despite a relatively high accuracy score on the training set (0.98), the accuracy score on the validation set is comparatively low (0.76). Therefore, overfitting was observed. Although we attempted to

mitigate this by reducing the number of trees and limiting the maximum depth of each tree, the model still struggled to generalize well. As a result, the validation accuracy remained suboptimal, making Random Forest a less effective choice for this specific task.

Model 5: Long Short-Term Memory Model

The Long Short-Term Memory network is a more advanced algorithm in our study. As a type of recurrent neural network, it is designed to handle sequential data and is particularly effective at capturing word order and contextual relationships. Thus, it is highly suitable for text classification tasks. Upon training, this model learns what words (or tokens in our case) are most important and are less important in affecting the category. That is, important tokens are tokens that, whenever appearing, cause a high probability of a certain class and a low probability of other classes. Less important tokens are tokens that have a similar conditional probability for all classes, and hence are not important in making classification. During actual classification, it remembers the important tokens and forgets the less important tokens upon reading the tokenized text, and makes classifications based on the remembered tokens.

```
230/230 ━━━━━━━━━━━━━━━━━━━━ 33s 125ms/step - accuracy: 0.7021 - loss: 1.6970 - val_accuracy: 0.7568 - val_loss: 0.8852
Epoch 2/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 41s 127ms/step - accuracy: 0.7480 - loss: 0.9306 - val_accuracy: 0.8299 - val_loss: 0.7773
Epoch 3/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 39s 116ms/step - accuracy: 0.8286 - loss: 0.7067 - val_accuracy: 0.8353 - val_loss: 0.7091
Epoch 4/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 41s 117ms/step - accuracy: 0.8450 - loss: 0.6036 - val_accuracy: 0.8299 - val_loss: 0.6792
Epoch 5/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 27s 119ms/step - accuracy: 0.8655 - loss: 0.5159 - val_accuracy: 0.8342 - val_loss: 0.6581
Epoch 6/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 41s 121ms/step - accuracy: 0.8793 - loss: 0.4653 - val_accuracy: 0.8484 - val_loss: 0.6282
Epoch 7/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 40s 119ms/step - accuracy: 0.8933 - loss: 0.4226 - val_accuracy: 0.8550 - val_loss: 0.6689
Epoch 8/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 42s 125ms/step - accuracy: 0.9124 - loss: 0.3503 - val_accuracy: 0.8495 - val_loss: 0.7120
Epoch 9/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 27s 119ms/step - accuracy: 0.9147 - loss: 0.3349 - val_accuracy: 0.8462 - val_loss: 0.6828
Epoch 10/10
230/230 ━━━━━━━━━━━━━━━━━━━━ 27s 118ms/step - accuracy: 0.9139 - loss: 0.3031 - val_accuracy: 0.8408 - val_loss: 0.6934
29/29 ━━━━━━━━━━━━━━━━━━━━ 1s 32ms/step
```

Figure 14: Accuracy Measure for Model 5

We implemented the LSTM model to evaluate its performance on our dataset. The results were impressive: the model achieved approximately 91% accuracy on the training set and nearly 84% on the validation set. This makes LSTM the best-performing model among all those tested in our project.

Conclusion

Model	**Training Accuracy**	**Validation Accuracy**
Naive Bayes	0.789	0.686
Logistic Regression	0.831	0.721
Linear SVM	0.984	0.746
Random Forest	0.980	0.758
LSTM	0.914	0.841

Table 2: Training and Validation Accuracy for Each Model

While the simpler models (Naive Bayes, Logistic Regression) showed moderate validation accuracy, the advanced models (Random Forest, Linear SVM) achieved high training accuracy ($> 95\%$) but suffered validation accuracy drops ($\sim$75%), suggesting overfitting. The LSTM model consistently performed best, achieving $\sim$91% training accuracy and $\sim$84% validation accuracy, effectively capturing contextual nuances in the text. One of the major issues is overfitting in Random Forest and SVM. As a response, we limited the tree depth, reduced the tree count, and adjusted SVM regularization, which decreased overfitting slightly but not fully. The success of the LSTM is significant due to its capacity to learn sequential language structures and long-range dependencies, whereas Random Forest

and SVM treat each token from each feature independently. "The assumptions that Naive Bayes and Logistic Regression rely on are unrealistic in this case; therefore, their performance is not as good as that of the LSTM."

Given additional time and resources, the first thing we would do is find a bigger dataset that has less missing labels. This not only can decrease the potential for overfitting but can also increase the performance of our models. We would also further tune the parameters used in Random Forest and SVM, either by experiments or learning from papers. In addition, we also want to use more data features like images or reviews to further enhance the performance of our model, if we can somehow convert information from images to inputable data. We are also interested in exploring some unsupervised learning approaches, such as clustering, to analyze our dataset and this research question. These methods might reveal different patterns and relationships in the data and subsequently give new insights beyond what was possible with our supervised approaches.